

VM DAILY ACCESS (SSH, VNC, EXPORT, PORTS)

Operators use the **Connect** hub on VM detail (platform and classic) for everyday tasks: graphical console, SSH, copy commands, guest ports, NAT exposure, and exports.

See also: Platform VM Detail UX (hero, action bar, Access tab layout).

Connect hub (platform)

On **VM detail ? Overview**, a single **Connect** card consolidates what used to be separate Daily access, laptop checklist, and inline NAT panels:

Section · Actions

Connect · **Open Cinema** ? ConsoleHub / noVNC; **SSH** ? in-browser terminal (hypervisor keys); copy guest IP and NAT-aware laptop `ssh` command

Laptop path · Inline checklist when the guest is on libvirt NAT (e.g. `192.168.122.x`): VM running ? guest IP ? expose SSH ? copy laptop command

Expose service · Presets (SSH, HTTP, HTTPS, ?) and custom port form; active NAT rules in a compact table (expand on Overview, full panel on **Access** tab)

Export · Download spec JSON, domain XML, both, or copy spec (shown on **Access** tab)

The card keeps `data-testid="vm-daily-access"` and `vm-laptop-access-checklist` for automation.

Access tab:

VM detail ? Access is the deep-work surface for connectivity:

- Full Connect hub with NAT panel expanded and export actions
- Links to **Guest security & ports** and **Guest health** tabs
- Same NAT presets as **Network ? Hypervisor NAT** and ConsoleHub **Network** lens

Use `?tab=access` to deep-link directly.

Header action bar:

The page header shows **Open Cinema**, **SSH**, one contextual power action (Start / Shutdown / Resume), and a **Power & more** overflow menu (pause, reboot, NMI, force stop, Studio, Virt-Viewer, pop out, delete). **Ask Zeus** opens Spotlight with VM blockers and suggested intents.

Attention stack:

Above the tabs, **at most one** banner is expanded (pending config shutdown, guest agent offline, or incomplete laptop NAT path). Lower-priority items appear as compact chips; dismiss persists per VM.

Classic VM detail

On classic **VM detail**, the Daily access strip below the power toolbar mirrors the Connect hub layout.

SSH from your laptop

- Machina does not store private keys.
- Cloud-init VMs are often **SSH-key only** ? use the private key that matches the injected public key: `ssh -i ~/.ssh/id_ed25519 -p 2222 ubuntu@HYPERVISOR_IP`.
- The UI prefers the cloud-init user (e.g. ubuntu) over stored SSH prefs when available.
- The UI copies hypervisor host + NAT port automatically when a rule exists (default SSH preset: host 2222 ? guest 22).
- Private guest IPs are **not** reachable directly from your laptop ? always connect via the hypervisor address and NAT port.

In-browser SSH (ConsoleHub Shell)

- Uses the hypervisor daemon's SSH keys, not your cloud-init key.
- When NAT is exposed, ConsoleHub dials 127.0.0.1:NAT_PORT on the hypervisor.
- If login fails, expose SSH and connect from your laptop with your private key.

VNC

- **Platform**: `/platform/vms/{id}/consolehub` ? Machine Cockpit with Display / Serial / Shell lenses.
- **Classic**: `/vms/{name}/console` ? same noVNC viewer.
- Linux cloud images often log in on **Serial** first; use ConsoleHub's Serial lens or the recovery card when password login is not configured.

Ports (platform)

Requires QEMU guest agent / Machina guest tools. Lists in-guest listeners via `GET /api/v1/zeus-firewall/vms/{id}/guest-ports`.

NAT port forwarding is managed on **VM detail** ? **Access** or **Network** ? Hypervisor NAT, in ConsoleHub **Network** lens, and inline from scanned ports / guest-access banners.

Known presets include SSH (2222?22), HTTP (9080?80), HTTPS (9443?443), databases, Grafana, etc. Custom named services can be saved per VM (server-side templates).

Platform Finder

In **Virtual Machines** column/list view, the inspector shows **VNC**, **SSH**, and **Copy IP** when a guest address is cached on the VM record.

API reference

Method · Path · Use

`GET` · /api/v1/vms/{id}/console` · VNC WebSocket path`

`GET` · /api/v1/vms/{id}/consolehub/plan` · ConsoleHub plan incl. guest_access, ssh_connect_host/port`

`GET` · /api/v1/vms/{id}/guest/health` · Guest IP, tools status`

`GET` · /api/v1/zeus-firewall/vms/{id}/guest-ports` · Listening ports`

`GET` · /api/v1/vms/{id}/port-forwards` · Active NAT rules for guest IP`

`POST` · /api/v1/vms/{id}/port-forwards` · Create NAT rule`

`GET/POST` · /api/v1/vms/{id}/port-forward-templates` · Custom service templates`

`GET` · /api/v1/vms/{id}/spec` · VM spec JSON`

`GET` · /api/v1/vms/{id}/domain-xml` · Live libvirt XML`

List VMs may include `guest_ip` and `guest_tools_status` when synced from health checks.

Automation

```
./scripts/e2e-vm-lifecycle-remote.sh user host --ssh-key ~/.ssh/id_ed25519
```

Verifies console API, guest ports, SSH login, and spec/XML export. Platform smoke tests cover NAT expose and ConsoleHub guest-access actions when VMs exist.

See also

- VM lifecycle and SSH keys